CHAPTER

10

Degraded DNA

DEGRADED DNA

DNA may be damaged or destroyed by adverse environmental conditions. Environmental exposure degrades DNA molecules by randomly breaking them into smaller pieces. Enemies to the survival of intact DNA molecules include water, oxygen, ultraviolent irradiation, and enzymes called nucleases that chew up DNA. These conditions are ubiquitous in nature and can result in fragmented DNA depending on the specific environmental situtation of a biological sample.

With older technologies such as restriction fragment length polymorphism (RFLP), severely degraded DNA samples would have been very difficult if not impossible to analyze as relative high molecular mass (aka molecular weight) DNA molecules needed to be intact in order to detect large VNTR (variable number of tandem repeat) alleles (Figure 10.1). The use of smaller short tandem repeat (STR) alleles enables better recovery of results with older, damaged DNA samples.

An ethidium-bromide stained agarose "yield gel" may be run to evaluate the quality of a DNA sample. Typically high molecular weight, high-quality genomic DNA runs as a relatively tight band of approximately 20,000 bp relative to an appropriate relative molecular mass marker. On the other hand, degraded DNA appears as a smear of DNA that is much less than 20,000 bp in size (Figure 10.2a).

Modern-day PCR methods, such as multiplex STR typing, are powerful because miniscule amounts of DNA can be measured by amplifying them to a level where they may be detected. Less than 1 ng of DNA can now be analyzed with multiplex PCR amplification of STR alleles compared to 100 ng or more that might have been required with RFLP only a few years ago. However, this sensitivity to low levels of DNA also brings the challenge of avoiding contamination from the police officer or crime scene technician who collects the biological evidence (see Chapter 4).

In order for PCR amplification to occur, the DNA template must be intact where the two primers bind as well as between the primers so that full extension can occur. Without an intact DNA strand that surrounds the STR repeat region to serve as a template strand, PCR will be unsuccessful because primer extension will halt at the break in the template.

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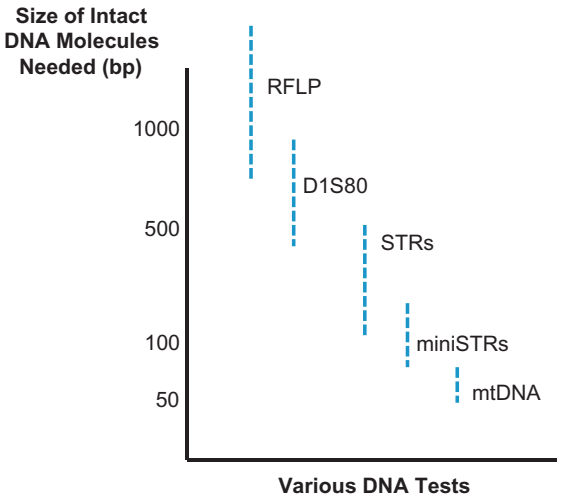


FIGURE 10.1 Illustration of required DNA fragment sizes for various DNA tests.

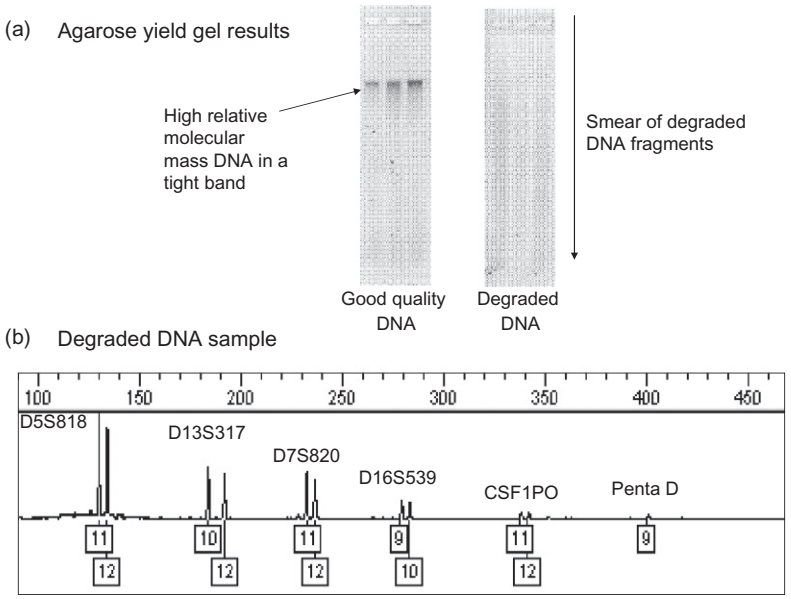


FIGURE 10.2 Impact of degraded DNA on (a) agarose yield gel results and (b) STR typing. (a) Degraded DNA is broken up into small pieces that appear as a smear on a scanned yield gel compared to good-quality DNA possessing intact high relative molecular mass DNA. (b) Signal strength is generally lost with larger-size PCR products when STR typing is performed on degraded DNA, such as is shown from the green dye-labeled loci in the PowerPlex 16 kit. Thus, 180 bp D13S317 PCR products have a higher signal than 400 bp Penta D amplicons because more DNA molecules are intact in the 200 bp versus the 400 bp size range.

The more degraded a DNA sample becomes, the more breaks occur in the template and fewer and fewer DNA molecules contain the full length needed for PCR amplification.

Benefits of STR Markers with Degraded DNA Samples

Fortunately, because STR loci can be amplified with fairly small product sizes, there is a greater chance for the STR primers to find some intact DNA strands for amplification. In addition, the narrow size range of STR alleles benefits analysis of degraded DNA samples because allele dropout via preferential amplification of the smaller allele is less likely to occur since both alleles in a heterozygous individual are similar in size.

A number of experiments have shown that there is an inverse relationship between the size of the locus and successful PCR amplification from degraded DNA samples, such as those obtained from a crime scene or a mass disaster (Whitaker et al. 1995, Sparkes et al. 1996, Takahashi et al. 1997, Schneider et al. 2004). The STR loci with larger-sized amplicons in a multiplex amplification, such as D18S51 and FGA or CSF1PO and Penta D, are the first to drop out of the DNA profile when amplifying extremely degraded DNA samples (Figure 10.2b).

In one of the first studies demonstrating the value of multiplex STR analysis with degraded DNA samples, the Forensic Science Service was able to successfully type a majority of 73 duplicate pathological samples obtained from the Waco Branch Davidian fire (see Chapter 9) with four STR markers (Whitaker et al. 1995). They observed no allele dropout and obtained concordant results on all samples where alleles were scored. A correlation was observed between successful typing at a locus and the average length of the alleles at that locus. The FES/FPS locus, which has alleles in the size range of 212 bp to 240 bp, only yielded 91 successful amplifications while the VWA locus with alleles ranging from 130 bp to 169 bp had 115 successful amplifications. Thus, loci with the larger alleles failed first. In addition, amelogenin amplicons (106 bp or 112 bp) were obtained on all 24 samples examined as part of the Waco identification program (Whitaker et al. 1995).

The potential for analysis of degraded DNA samples is an area where multiplex STR systems really shine over previously used DNA markers. STRs are more sensitive than single-locus probe RFLP methods, less prone to allelic dropout than VNTR systems (AmpFLPs) such as D1S80, and more discriminating than other PCR-based typing methods, such as HLA-DQA1 and AmpliType PolyMarker (see Butler 2010, *Fundamentals*, Chapter 3). Nevertheless, either due to degradation or inhibition, partial STR profiles still occur (Figure 10.3).

miniSTRs

The Use of Reduced Sized PCR Products (miniSTRs)

In an article entitled "Less is more – length reduction of STR amplicons using redesigned primers," Wiegand and Kleiber (2001) demonstrated that highly degraded DNA as well as very low amounts of DNA could be more successfully typed using some new redesigned PCR primers that were close to the STR repeat compared to the established sequences that generated longer amplicons for the same loci. This work built upon some previous work with creating smaller PCR products for mass spectrometry. Table 10.1 provides a timeline

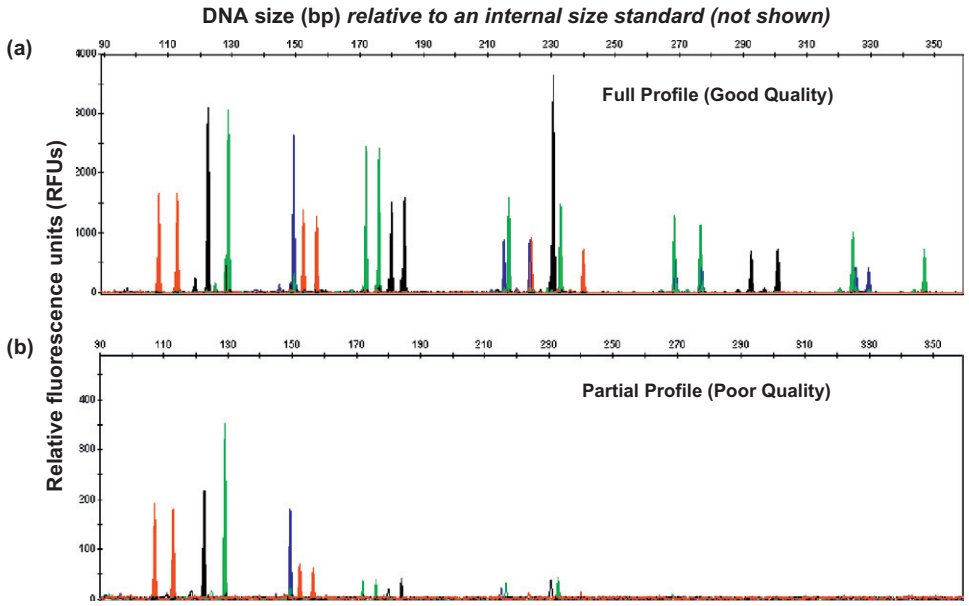


FIGURE 10.3 A comparison of DNA profiles originating from the same biological source but of different qualities. (a) Intact, good-quality DNA yields a full profile. (b) Degraded, poor-quality DNA yields a partial profile with only the lower-size PCR products producing detectable signal. With the degraded DNA sample shown in (b), information is lost at the larger-sized STR loci. Also note the lower relative fluorescence units with the poor-quality partial profile in (b). Figure courtesy of Margaret Kline, NIST.

of the history surrounding the development and implementation so-called "miniSTRs" or reduced-sized PCR products.

STR loci used in commercially available kits can extend past 400 bp in size. Most of this length comes from the flanking sequences surrounding the STR repeat of interest. PCR primers for larger-sized STR markers have been moved away from the repeat region that imparts variability to the locus in order to fit into a desired size range for a particular multiplex assay (e.g., Krenke et al. 2002).

For example, the two PCR primers used for the PowerPlex 16 locus Penta D anneal 71 bp upstream and 247 bp downstream of the core AAAGA repeat. Amplification with these PCR primers generates amplicons in the size range of 376 bp to 449 bp with alleles ranging from 2.2 to 17 repeats (Krenke et al. 2002). When primers are brought to within 11 bp upstream and 19 bp downstream of the repeat region, the overall PCR product sizes drop by 282 bp to a range of 94 bp to 167 bp for alleles 2.2 bp to 17 bp (Butler et al. 2003). Figure 10.4 illustrates this size reduction principle when creating reduced-size STR amplicons or "miniSTRs." It is important to keep in mind that some loci can be reduced in size more than others (Table 10.2).

Several disadvantages do exist for miniSTRs. A major disadvantage is that only a few loci can be simultaneously amplified in a multiplex because the size aspect has been removed. Large multiplex assays like PowerPlex 16 pack four or more loci into a single dye color by shifting primers away from the repeat region to make larger PCR products. The "miniplexes" created for amplifying miniSTRs have primers that are as close as possible to the

TABLE 10.1 Timeline of Events Surrounding Development and Use of miniSTR Loci for Forensic DNA Typing (see http://www.cstl.nist.gov/biotech/strbase/miniSTR/timeline.htm).

Date	Event*
1994	In running degraded DNA samples from remains of victims of the Branch Davidian fire in Waco, Texas, the Forensic Science Service find that smaller STRs in their 4plex work better than the larger loci (Lygo et al. 1994, Whitaker et al. 1995).
1996	GeneTrace Systems, a company in Menlo Park, California, begins work with shorter STR amplicons to perform rapid DNA typing using MALDI-TOF mass spectrometry; small PCR products are necessary to obtain successful results with MALDI-TOF.
Sept 1997	John Butler and coworkers from GeneTrace submit a patent on mass spec typing and multiplexing using small STR amplicons; U.S. patent 6,090,558 is granted July 18, 2000 (Butler et al. 2000).
1997–1999	Butler and coworkers from GeneTrace (funded by National Institute of Justice Grant 97-LB-VX-0003) give numerous talks on TOF-MS STR typing with smaller PCR products; some of their work is published (Butler et al. 1998a, Butler 1999); manuscripts are also published as part of the International Symposium on Human Identification in 1997 (Butler et al. 1998b) and 1998 (Butler & Becker 1999) as well as the Second European Symposium on Human Identification in 1998 (Butler et al. 1999).
June 1998	Hermann Schmitter of the German BKA hears Kathy Stephens from GeneTrace speak on small STRs and independently decides to try them on degraded DNA samples; the efforts of his laboratory lead to work with STR typing of telogen hair shafts (Hellmann et al. 2001).
Summer 2000	Butler (now back at NIST) puts fluorescent dyes on his GeneTrace mass spec primers and demonstrates reliable STR typing using the ABI 310; this work is presented as a poster at the 11th International Symposium on Human Identification (Ruitberg & Butler 2000).
Summer 2001	Bruce McCord (Ohio University) joins John Butler in an informal collaboration working on short STRs to aid analysis of degraded DNA samples.
October 2001	The National Institute of Justice publishes Butler's final report from his work at GeneTrace Systems including primer sequences for the short STR amplicons (see p. 24, Butler & Becker 2001).
November 2001	After discussions with the World Trade Center Kinship and Data Analysis Panel (WTC KADAP), Robert Shaler, of the New York City Office of the Chief Medical Examiner (OCME), who is leading the efforts to identify victims of the WTC attacks using DNA testing, contacts Butler at NIST and asks that efforts with short STRs be accelerated so that this technology may be used with aiding WTC victim identifications; the term "miniSTR" is coined at this time.
December 2001–January 2002	Butler, McCord, and postdoctoral student Yin Shen work to develop five miniplex assays that incorporate all of the CODIS STR loci as well as D2S1338, Penta D, and Penta E; some concordance studies and initial validation studies are performed but the primary focus is on primer development, generation of allelic ladders and creation of Genotyper macros rather than assay optimization in terms of sensitivity; a summary of this work is later published (Butler et al. 2003).
January 16, 2002	Butler speaks at SWGDAM (Quantico, VA) informing the attendees of research progress with miniSTRs; Mitch Holland of Bode Technology Group is present and learns of this effort; Butler meets with Shaler and Howard Baum of NYC OCME to make decisions about which miniplex would be tried by NYC OCME.
February 2002	"Big Mini" assay and protocols are supplied to NYC OCME and New York State Police in Albany for trial use; OCME work with Big Mini proves sensitive down to 0.5 ng of DNA template with the provided protocol of 28 cycle PCR and 1 U of TaqGold in 25 μL volumes.

(Continued)

TABLE 10.1 Timeline of Events Surrounding Development and Use of miniSTR Loci for Forensic DNA Typing (see http://www.cstl.nist.gov/biotech/strbase/miniSTR/timeline.htm). *(Continued)*

Date	Event*
February 22, 2002	Butler speaks to the WTC KADAP as part of their meeting in Albany, NY to inform the group of research progress with miniSTRs.
March 2002	McCord and Butler submit a proposal to the National Institute of Justice to further develop miniSTR assays; NIJ Grant 2002-IJ-CX-K007 is awarded and results in several publications (Chung et al. 2004, Drábek et al. 2004, Opel et al. 2006, Opel et al. 2007, Opel et al. 2008).
March 2002	Shaler meets with Jim Schumm of the Bode Technology Group and discusses the possibility of Bode performing miniSTR testing; Bode is already testing bones from the WTC investigation with conventional STR methods.
April 4, 2002	Schumm contacts Butler for primer sequences and information on miniplex work performed to date; an Excel file and PowerPoint presentation are emailed from NIST to Bode to aid their efforts.
Summer 2002	BodePlexes are developed by Schumm and coworkers at the Bode Technology Group using NIST supplied information as well as the NIJ-published GeneTrace report; a majority of the PCR primers are kept the same but dye labels are switched to provide different marker combinations; BodePlexes utilize 5X amount of TaqGold and 4–6 more PCR cycles to improve sensitivity over previous miniSTR work.
November 22, 2002	Schumm invites Shaler, Baum, and Butler to visit the Bode Technology Group (Springfield, VA) in order to review their validation studies for the BodePlex 1 and BodePlex 2 assays that were intended for use on WTC samples; while PCR primer sequences were not revealed, it was acknowledged that a majority of them came from the NIST and GeneTrace information.
Late 2002–Summer 2003	BodePlex 1 (D13S317, D21S11, D7S820, D16S539, and CSF1PO) and BodePlex 2 (TPOX, FGA, D7S820, and D18S51) are run by the Bode Technology Group on the bone and tissue samples that are part of the WTC investigation—the use of miniSTR BodePlexes has been described in several publications (Holland et al. 2003, Schumm et al. 2004).
September 2003	Butler, Shen, and McCord's publish miniSTR primer sequences (Butler et al. 2003) for 13 CODIS STRs and miniplex sets developed in late 2001 and early 2002.
December 2003	Mike Coble begins work at NIST as an NRC postdoctoral associate to develop miniSTRs for loci beyond the CODIS core.
February 2004	Coble speaks at the American Academy of Forensic Sciences meeting in Dallas, TX about efforts at NIST to develop new miniSTR assays. Denise Chung, a graduate student from Bruce McCord's lab at Ohio University, speaks on their progress with miniSTRs.
April 2004	Butler is invited to participate in the European DNA Profiling Group (EDNAP) and European Network of Forensic Science Institutes (ENFSI) meeting in Cyprus to help setup an interlaboratory test with Peter Gill on degraded DNA; this EDNAP/ENFSI study will involve conventional STR testing, miniSTR assays, and SNP assays on the same degraded DNA samples.
July 2004	Publication of McCord's lab work on miniSTRs (Chung et al. 2004, Drábek et al. 2004).
September 2004	EDNAP/ENFSI study involving miniSTRs and SNPs is initiated—materials for two miniSTR systems are provided by NIST to all participating laboratories: see http://www.cstl.nist.gov/biotech/strbase/miniSTR.htm#Protocols.
January 2005	Coble publishes his work with new miniSTR loci (Coble & Butler 2005).
September 2005	Coble outlines work with new miniSTR loci at the International Society of Forensic Genetics meeting (Coble et al. 2006).

(Continued)

TABLE 10.1 Timeline of Events Surrounding Development and Use of miniSTR Loci for Forensic DNA Typing (see http://www.cstl.nist.gov/biotech/strbase/miniSTR/timeline.htm). *(Continued)*

Date	Event*
October 2005	Applied Biosystems announces at the ENFSI meeting that they are working on a commercial kit involving miniSTR loci, which should be available in 2006.
January 2006	Gill, Lyn Fereday, Niels Morling, and Peter Schneider representing the EDNAP and ENFSI groups propose that miniSTRs be adopted as the way forward to increase both the robustness and sensitivity of forensic DNA analysis. Their recommendations are published (Gill et al. 2006). They also recommend that three new miniSTR loci be adopted by European laboratories: D10S1248, D14S1434 (now replaced by D2S441), and D22S1045—all loci described by Coble & Butler (2005).
February 2006	NIST staffer Becky Hill presents a poster at the AAFS meeting in Seattle, WA on allele frequencies in U.S. populations found with 27 new miniSTR loci under development at NIST (Hill et al. 2006a).
February 2006	Applied Biosystems announces at the AAFS meeting in Seattle, WA efforts to develop a 9plex miniSTR kit (using their 5-dye and mobility modifier technology); this miniSTR kit is designed to recover information from the larger loci in their Identifiler kit and will amplify the following loci: D13S317, D7S820, D2S1338, D21S11, D16S539, D18S51, CSF1PO, FGA, and amelogenin.
March 2006	McCord's group publishes their work using miniSTRs on skeletal remains (Opel et al. 2006).
October 2006	Hill presents a poster at the Promega meeting in Nashville, TN on characterization of 26 miniSTR loci under development at NIST (Hill et al. 2006b).
October 2006	Applied Biosystems publishes information about their new MiniFiler kit in their marketing newsletter Forensic News.
March 2007	Applied Biosystems releases their new MiniFiler kit.
July 2007	NIST and Applied Biosystems co-publish an article examining concordance of allele calls between the MiniFiler kit and other commercial STR kits (Hill et al. 2007).
July 2008	Applied Biosystems publishes their developmental validation work on MiniFiler (Mulero et al. 2008).

**Some additional work with small PCR products for STRs has been published that is not included on this timeline (see reference list at the back of the chapter).*

repeat region and therefore typically only have one locus per dye color because all of the loci have about the same size range of ≈100 bp (Butler et al. 2003). However, using mobility modifiers to adjust the electrophoretically observed PCR product sizes, Applied Biosystems was able to put eight miniSTRs and amelogenin into their single amplification MiniFiler kit (Mulero et al. 2008) (Figure 10.5).

Since different PCR primers are in use with miniSTRs compared to conventional STR megaplexes, it is important that concordance studies be performed to verify that allele dropout from primer binding site mutations is rare or non-existent (Hill et al. 2007). This is performed by examining the genotyping results to see if they are the same between the primer sets (see Chapter 6). Occasionally a point mutation or an insertion or deletion may occur in the flanking region outside of a miniSTR primer binding site which can lead to a problematic (and undetectable) difference in a heterozygous allele call (Butler et al. 2003, Drabek et al. 2004, Hill et al. 2007).

Regardless of these disadvantages, it is likely that miniSTRs will play a role in the future of degraded DNA analysis, helping recover information on larger loci that has been lost using conventional megaplex amplification.

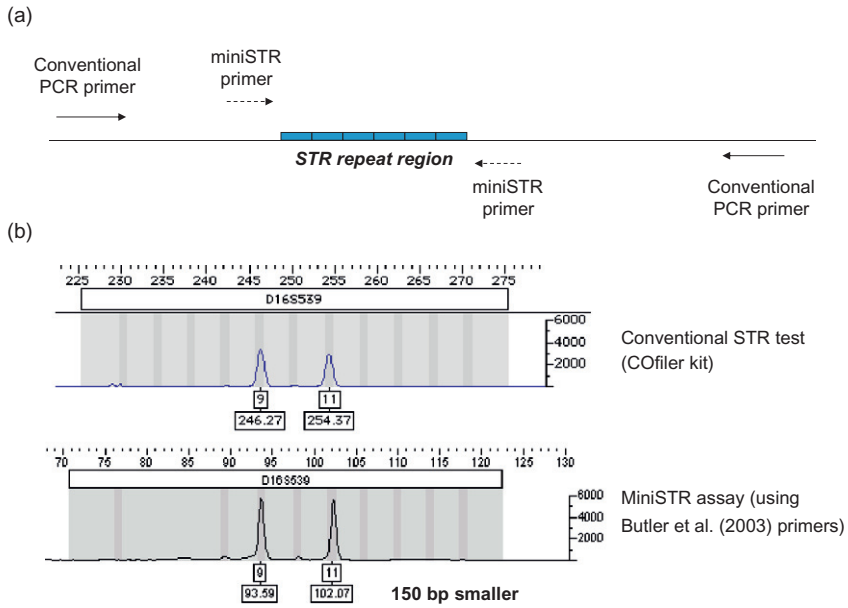


FIGURE 10.4 (a) MiniSTRs or reduced-sized amplicons for STR typing are created by designing PCR primers that anneal closer to the repeat region than conventional STR kit primers. (b) PCR product sizes, such as demonstrated here with D16S539, can be reduced by over 150 bp relative to conventional tests. MiniSTR assays can produce the same typing result as those from larger STR amplicons produced by kits often with greater success on degraded DNA samples.

With enzyme-digested DNA, miniSTR loci performed better than loci from a commercial STR kit (Chung et al. 2004). As described in Chapter 9, reduced-size STR assays helped make possible some of the World Trade Center victim identifications from burned and damaged bone samples (Schumm et al. 2004). Even telogen hair shafts, which contain very little nuclear DNA, have been successfully typed using reduced-size STR amplicons (Hellman et al. 2001, Müller et al. 2007, Opel et al. 2008).

In 2006, the European DNA Profiling Group (EDNAP) published the results of a study with degraded DNA samples where miniSTR primer sets were compared to conventional STR multiplex kits and an experimental single nucleotide polymorphism (SNP) assay (Dixon et al. 2006). Overall, the miniSTR assays performed the best on the degraded DNA samples, which led leaders of the European community to advocate for miniSTR loci in future STR kits (Gill et al. 2006).

New miniSTR Loci

STR loci other than the CODIS markers and others that are currently used in forensic DNA typing have been developed as miniSTR systems with a focus on loci possessing small alleles and narrow size ranges (Ohtaki et al. 2002, Coble & Butler 2005, Hill et al. 2008).

TABLE 10.2 PCR Product Size Reduction Obtained with New Primers for Several miniSTR Loci (Butler et al. 2003).

Locus	miniSTR Size (bp)	Allele Range (repeat numbers)	Size Reduction*
TH01	51–98	3–14	−105 bp
TPOX	65–101	5–14	−148 bp
CSF1PO	89–129	6–16	−191 bp
VWA	88–148	10–15	−64 bp
FGA	125–281	12.2–51.2	−71 bp
D3S1358	72–120	8–20	−25 bp
D5S818	81–117	7–16	−53 bp
D7S820	136–176	5–15	−117 bp
D8S1179	86–134	7–19	−37 bp
D13S317	88–132	5–16	−105 bp
D16S539	81–121	5–15	−152 bp
D18S51	113–193	7–27	−151 bp
D21S11	153–211	24–38.2	−33 bp
Penta D	94–167	2.2–17	−282 bp
Penta E	80–175	5–24	−299 bp
D2S1338	90–142	15–28	−198 bp

**Compared to Applied Biosystems STR kits except for Penta D and Penta E, which are in Promega's PowerPlex 16 kit.*

A battery of additional assays has been made available to aid researchers and forensic practitioners in the future when working with degraded DNA specimens.

Scientists at the U.S. National Institute of Standards and Technology (NIST) selected a set of 26 new miniSTR loci from over 900 candidate STR loci based on expected allele ranges and the ability to design PCR primers close to the flanking region (Coble & Butler 2005, Coble et al. 2006). These 26 loci were characterized in U.S. population samples (Table 10.3). In addition, 25 of the loci were combined with amelogenin into a single 26plex assay for typing reference samples (Hill et al. 2009).

Nucleosome Protected Areas of the Genome

Another approach to enabling better recovery of results from potentially damaged DNA samples is to look for regions of the DNA that are better protected when DNA is chromosomally packaged inside the cell. In theory, when DNA is bound to histone proteins in its chromosome packaging, there will be less opportunity for degradative enzymes, such as

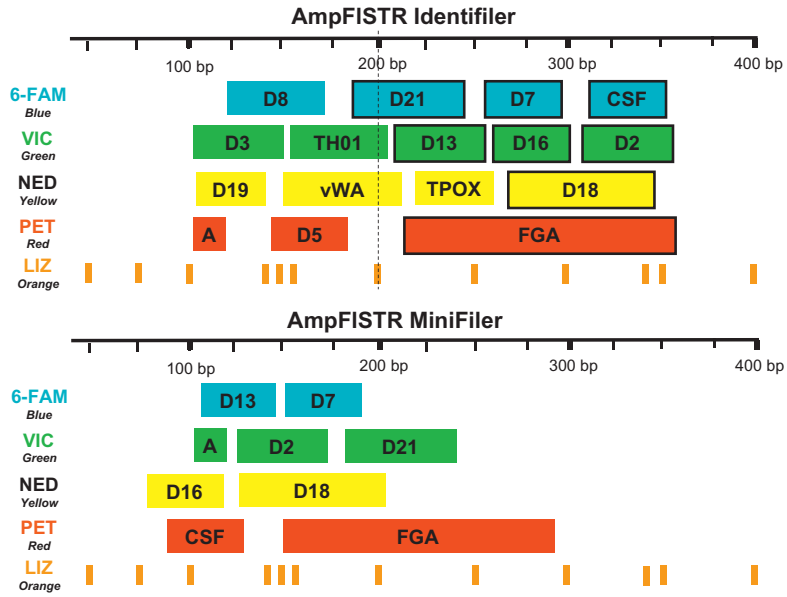


FIGURE 10.5 An illustration of the relative sizes and dye labels for PCR products generated with the Identifiler and MiniFiler STR kits. MiniFiler is designed to recover the larger sized Identifiler PCR products (those greater than 200 bp, with the exception of TPOX, as shown by the dotted line).

DNase I, to break the DNA into smaller fragments. The nucleosome core particle complex generally protects 147 bp regions of a DNA sequence (Richmond & Davey 2003). Efforts have been made to discover the predominant DNA sequences found within these nucleosome core particles in order to predict how well DNA sequences would be expected to survive DNA damage (Ioshikhes et al. 2006, Radwan et al. 2008).

Researchers at the University of Strathclyde in Glascow, Scotland have performed *in silico* studies on 60 forensic DNA markers, 58 STRs plus amelogenin X and Y sequences, in search of properties that promote or inhibit the DNA sequence from binding to histones (Thanakiatkrai & Welch 2010). They found that the majority of STR loci commonly used are likely to be bound as nucleosomes and thus better protected against DNA damage. They also propose that selection of markers, which appear to be protected best, might be used in the future to create an improved multiplex assay for degraded DNA samples.

The approach of finding optimal protected regions of DNA in order to design assays that will improve success rates on forensic casework samples is still in its infancy. It may be that different DNA sequence regions will work better than the STR loci in use today. However, the large national databases around the world containing millions of STR profiles (see Chapter 10) will probably discourage movement to other DNA markers unless success rates on casework samples with degraded DNA are significantly improved. Only time will tell if this approach plays a significant role in the future of forensic DNA typing.

TABLE 10.3 Information on 26 miniSTRs Characterized at NIST (Hill et al. 2008). Chromosomal Location and Base Pair (bp) Position of Each Marker Was Determined by Using BLAT (http://genome.ucsc.edu/cgi-bin/hgBlat) and the May 2004 Assembly of the Human Genome. The Observed Size Range (bp) Values Are Based on the miniSTR Amplicon Sizes That Were Observed in This Study Relative to the GS500 LIZ Size Standard. The Loci Are Listed in Order, Highest to Lowest, by the Overall Heterozygosity Values.

Locus	Repeat Motif	Chromosomal Location	Chromosome Position	Observed Size Range (bp)	N	Overall Heterozygosity
D9S2157	ATA	9q34.2	Chr 9 133.065 Mb	71–101	661	0.844
ATA63 (D12)	YAA	12q23.3	Chr 12 106.825 Mb	76–106	659	0.829
D10S1248 (NC01)	GGAA	10q26.3	Chr 10 130.567 Mb	83–123	663	0.792
D22S1045 (NC01)	ATT	22q12.3	Chr 22 35.779 Mb	76–109	663	0.784
D2S441 (NC02)	TCTA	2p14	Chr 2 68.214 Mb	78–110	660	0.774
D10S1435	TATC	10p15.3	Chr 10 2.233 Mb	82–139	663	0.766
D2S1776	AGAT	2q24.3	Chr 2 169.471 Mb	127–161	654	0.763
D3S4529	ATCT	3p12.1	Chr 3 85.935 Mb	111–139	660	0.761
D6S474	GATA	6q21	Chr 6 112.986 Mb	107–135	648	0.761
D5S2500	GRYW	5q11.2	Chr 5 58.735 Mb	85–125	664	0.747
D1S1627	ATT	1p21.1	Chr 1 106.676 Mb	81–100	660	0.746
D1S1677 (NC02)	TTCC	1q23.3	Chr 1 160.747 Mb	81–117	660	0.746
D6S1017	ATCC	6p21.1	Chr 6 41.785 Mb	81–109	664	0.740
D3S3053	TATC	3q26.31	Chr 3 173.234 Mb	84–108	648	0.739
D9S1122	TAGA	9q21.2	Chr 9 76.918 Mb	93–125	659	0.734
D17S974	CTAT	17p13.1	Chr 17 10.459 Mb	95–123	664	0.732
D11S4463	TATC	11q25	Chr 11 130.338 Mb	88–116	664	0.730
D4S2408	ATCT	4p15.1	Chr 4 30.981 Mb	85–109	654	0.722
D18S853	ATA	18p11.31	Chr 18 3.981 Mb	82–103	664	0.711
D20S1082	ATA	20q13.2	Chr 20 53.299 Mb	73–100	664	0.696
D14S1434 (NC01)	CTRT	14q32.13	Chr 14 93.298 Mb	70–98	663	0.696
D20S482	AGAT	20p13	Chr 20 4.454 Mb	86–126	648	0.691
GATA113 (D1)	GATA	1p36.23	Chr 1 7.377 Mb	81–105	654	0.668
D8S1115	ATT	8p11.21	Chr 8 42.656 Mb	63–96	664	0.663
D17S1301	AGAT	17q25.1	Chr 17 70.193 Mb	114–138	664	0.649
D4S2364 (NC02)	ATTC	4q22.3	Chr 4 93.976 Mb	67–83	660	0.511

ATTEMPTS WITH DNA REPAIR FOLLOWING DNA DAMAGE

New England Biolabs (Ipswich, MA) sells an enzyme cocktail to help repair DNA damage prior to performing PCR. While their "PreCR Repair Mix" cannot repair fragmented DNA, it contains a ligase and other enzymes capable of repairing nicks, thymine dimers caused by UV radiation, and abasic sites caused by hydrolysis. The effectiveness of the PreCR Repair Mix on artificially damaged DNA and on non-probative casework samples was examined by researchers at the Alcohol, Tobacco, and Firearms (ATF) and Armed Forces DNA Identification Laboratories (Bille et al. 2008), but unfortunately their experiments did not show any major advantages with the PreCR Repair Mix on the samples they tested.

The National Institute of Justice (NIJ) has funded several projects to work on repair of damaged DNA templates (Ballantyne 2006, Nelson 2009) and some promising results have been obtained. New DNA polymerases capable of PCR amplification from damaged or ancient DNA samples have been developed and studied (McDonald et al. 2006, d'Abbadie et al. 2007, Shapiro 2008).

John Nelson of the GE Global Research Center (Niskayuna, NY) in his NIJ final report (Nelson 2009) concludes that DNA may be damaged in a number of ways and not all of them can be repaired. He notes that "there is likely a 'window of opportunity' during exposure of DNA to sub-optimal conditions when the DNA molecular weight is large enough to still contain segments of DNA that encompass STR loci, but contain lesions that can be repaired" (Nelson 2009). He found that with the addition of enzymes such as Fpg (formamidopyrimidine [fapy]-DNA glycosylase), *E. coli* endonuclease IV, *E. coli* endonuclease VIII, and T4-PDG (pyrimidine dimer glycosylase) to a DNA polymerase/ligase enzyme cocktail there was an increase in the ability to repair DNA. Unfortunately, the use of additional enzymes means that there is a greater potential for introducing contaminating DNA in these reagents. Therefore, the enzyme cocktail reagents would have to be carefully screened by manufacturers to avoid contaminating forensic casework samples as well as being checked by forensic laboratories with additional reagent blanks to confirm that spurious DNA was not added (see D.N.A. Box 4.3).

Perhaps future DNA repair kits will enable laboratories to obtain results from forensic casework samples that currently fail to amplify or produce partial profiles.

READING LIST AND INTERNET RESOURCES

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